



How do digital startups manage their activities? Insights and opportunities for business process management

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Abstract

Business process management (BPM) aims to help organizations manage their business processes. Startups differ from established firms as they go through different phases of prospecting, developing, and exploiting the new venture. Startups begin to focus on the organization of their processes after they reach the exploiting (scale-up) phase. Digital startups are unique as information technology (IT) becomes the business model itself. These unique characteristics raise a question: how do digital startups at the scale-up phase manage their business processes? To answer the question, two case studies on digital startups in logistics providers are conducted. The case studies are designed to be inductive in nature. Grounded Theory Method (GTM) is used for data collection and analysis. Data is collected via interviews and supporting documents. The BPM capability provides the basis to create guiding questions for the interviews. The interview results are analyzed with a grounded theory approach of open, theoretical, and selective coding. To derive a new theory, cross-case analyses are conducted. Findings from two digital startups allow us to identify important categories that play a role in how digital startups manage their activities: industry and stakeholders, digital offerings, organic structure, process management, performance measurement, employee training and culture. We further theorize that the competitive nature of startups makes them customer-centric and focus on agility. Digital startups continuously improve their product and conduct adaptive process experimentation involving a cycle of process identification, IT-based process implementation and process adaptation. The supporting capabilities that enable the process management of digital startups are agile people and culture and organic structure.

Keywords Digital startups · Digital entrepreneurship · Capability · Business process · Business process management

1 Introduction

Every organization conducts a series of interrelated activities to achieve its goals known as business processes (Van Looy et al. 2014). The organization and management of business processes are the focus of the Business Process Management (BPM) discipline. BPM has been applied in traditional organizations to help them identify, discover, analyze, redesign, implement, and monitor their business processes known as the BPM lifecycle.

The recent development in IT drove the emergence of digital entrepreneurship or digital startups. Startups are a specific type of new venture creation (NVC) that has been addressed in the entrepreneurship domain. The wave of information technology (IT) development greatly affects entrepreneurial processes and drives research on IT-associated startups, or digital entrepreneurship (DE). According to Steininger (2019), there are different ways in which IT plays a role in supporting entrepreneurial operations and value creation. IT-associated entrepreneurship is classified as (1) a *facilitator* when IT facilitates making the startup's operations easier, (2) a *mediator* when it mediates new ventures' operations, (3) an *outcome* of entrepreneurial operations, and (4) a *digital startup* when IT plays a role as ubiquity and becomes the business model itself.

Startups differ from established enterprises because they go through several entrepreneurial stages, including prospecting, developing, and exploiting (Bakker and Shepherd 2017; Davidsson and Gruenhagen 2021; von Briel et al. 2018). As entrepreneurial organizations grow, they transition from practices that enable them to thrive in the startup phase to mature, professional practices involving standardization and delegation of authority (Tumbas et al. 2017). Startups evolve and change as they adjust, reflect, and re-invent their processes and value propositions. As startups reach the exploitation or scale-up phase, they establish processes, albeit not as structured or standardized as traditional organizations.

BPM as a discipline is often criticized for its inability to deal with continuous and significant change and uncertainty (Szelągowski 2019; Badakhshan et al. 2020), implying that classic BPM is difficult to apply to startups in changing conditions. Despite these challenges, several works have investigated the implementation of BPM in the context of startups. de Santos and de Pádua (2023) argued BPM is a management approach that can bring the development of dynamic capabilities (DC) to start-ups. They developed a BPM promotion framework that operationalizes the development of DCs in startups. However, they do not focus on how startups manage their activities. Griva et al. (2023) focus on defining growth in growing early-stage startups and identifying their capabilities, which include culture, learning organization, adaptability and selling capabilities.

Previous studies do not focus specifically on the context of digital startups that have reached their exploiting (scale-up) phase. In digital startups, IT acts as support, product and business model. Thus, the way they manage their activities and processes may differ from other types of IT-associated startups. Other works from different domains have addressed different startup capabilities, including people (El Hanchi and Kerzazi 2020) and culture, team/people and method (de Oliveira Mota

et al. 2021; Griva et al. 2023), but not as interrelated capabilities to manage business processes. Despite the unique characteristics of digital startups, once they reach the scale-up phase, they must have interrelated activities that must be managed to achieve their goals. This leads us to a research question that needs to be addressed: *How do digital start-ups that have already reached the exploitation (scale-up) phase manage their business processes?*

To answer the question, we used the BPM capability perspective and conducted two case studies on digital startups in logistics providers. As Davidsson and Gruenhagen (2021) stated, obtaining cases from startups is challenging because of their ever-changing conditions. Thus, our criteria for choosing these startups are mainly due to willingness to be involved in the research. Data collection and analysis were conducted with grounded theory methodology (GTM). Data are mainly collected from interviews and supported by documents. The analysis includes open, theoretical and selective coding.

This paper's main contribution is providing insights into how startups in the scale-up phase manage their activities using the BPM capability as a theoretical lens. These findings enrich the BPM literature regarding process management in the context of digital startups. At the same time, this paper extends the current NVC literature with more detailed insights into how startups manage their activities. From a practical point of view, the findings provide insights for digital entrepreneurs into some important factors they need to consider in managing their activities.

This paper is structured as follows. Section 2 reviews previous works on BPM capability, the driving force to break away from the classical BPM, and BPM in startups that provide a foundation for this research. We then describe our research method in Sect. 3. Section 4 describes the results from the case studies where we describe how logistics digital startups manage their activities and the proposed theoretical model. Then, we discuss the theoretical implications of our findings for BPM and new venture creation (NVC) literature, opportunities for future study, and limitations of our study in Sect. 5. Concluding remarks are provided in the final section.

2 Background

2.1 BPM capability

Business process management (BPM) has received much attention as a discipline that helps organizations manage their business processes. Over the last decades, BPM researchers have developed frameworks, methods, and tools to support process-related management activities and help organizations design and improve their business processes. One key notion is that the management of business processes must go through continuous phases of process identification, discovery, analysis, redesign, and monitoring, known as the business process lifecycle (Dumas et al. 2021).

The focus in the development of the BPM discipline has shifted from developing methods and tools to increasingly focusing on the advancement of BPM capability in organizations, i.e., the skills and routines to execute business processes and

employ BPM methods and tools for incremental and radical process change (Niehaves et al. 2014; Poeppelbuss et al. 2015).

To implement process orientation in organizations, successful BPM requires capability related to the core elements of BPM: Strategic Alignment, Governance, Methods, IT, People, and Culture (Rosemann and vom Brocke 2015). Method and IT-related capability areas are commonly structured according to the phases of the BPM lifecycle: process design, implementation, execution, monitoring, and improvement.

The six core elements of BPM have been used to develop maturity models, a tool to help organizations assess how well they are managing their business processes (Van Looy et al. 2014, 2021). The maturity model assumes certain progression on how established firms improve their BPM capabilities ((Röglinger et al. 2012; Tarhan et al. 2016). With the current business environment, the context of organizations varies and the way they manage their processes might differ accordingly.

Van Looy et al. (2014) elaborated on BPM capability by providing metrics to measure and assess the management of business processes in an organization. They proposed four main BPM capability areas, namely life cycle, management, culture, and structure. The life cycle capability area represents the different stages of the process lifecycle, including Plan, Do, Check, and Action (PDCA). The next area is management capability, comprising strategy to assess how far the organization's goals are used to manage business processes, external relationships to assess the relationship with stakeholders outside the organization, roles to assess how well the team's roles and relationships within the organization, skills to assess the organization's ability to achieve goals and face all challenges in business. The next capability is culture, which consists of values, attitudes and behaviors, appraisals, and top management. The final capability is the organizational structure, which considers the organizational chart and the existence of governance bodies. In this study, we use the BPM six core elements and BPM capability as the basis for developing the guiding questions for the data collection of our study.

2.2 Breaking the classical BPM

Many researchers cautioned that BPM is not a one-size-fits-all solution and that it must consider different contextual factors (vom Brocke et al. 2016, 2021). vom Brocke et al. (2016) describe the BPM context factors relevant to BPM projects based on their settings. The model helps characterize a BPM initiative according to its goals, the process's characteristics, and the organization's and external environment's characteristics. The organizational dimension, which includes the organization's industry, size, and culture, determines the right BPM approach. Regarding organizational size, vom Brocke et al., differentiate startups from small and medium enterprises (SMEs) and large enterprises. This differentiation acknowledges that a startup is a unique type of organization that cannot be treated the same as an established organization.

von Rosing et al. (2015) emphasize the need for flexibility and change in using BPM. Several studies stated that classical BPM is no longer sufficient in a rapidly

changing environment due to digital transformation (Mendling et al. 2020; Baiyere et al. 2020). Baiyere et al. (2020) argued that the three logics of BPM—modeling (process), infrastructural alignment (infrastructure), and procedural actor (agency) need to be updated in a company undergoing digital transformation. They proposed new logic: light touch processes (process), infrastructure flexibility (infrastructure), and mindful actors (agency).

Badakhshan et al. (2020) argue for more contingent approaches to BPM that the business process needs to align with external dynamics. These arguments call for an agile BPM. They describe agile BPM as “...make BPM continuously ready to create change rapidly or inherently, embrace change proactively or reactively, and learn from change to facilitate consistent outcomes of business processes and contribute to business performance and perceived customer value (economy, quality, and simplicity)” (Badakhshan et al. 2020, p. 11). The authors argue that such capability can be done by “continuously codifying and integrating requirements, and understanding, documenting, designing, analyzing, simulating, executing, and monitoring business processes” (Badakhshan et al. 2020, p.3).

2.3 Startups and BPM

2.3.1 Startups as new venture creation and digital entrepreneurship

Discussion about startups has been addressed in entrepreneurship and new venture creation (NVC) domains. Much attention in the entrepreneurship literature is given to the entrepreneurial process. According to Shane (2003), the entrepreneurial process involves the identification and evaluation of opportunity, the decision on whether to exploit it, the efforts to obtain resources, the process of organizing those resources into a new combination, and the development of a strategy for the new venture.

Other works have attempted to address the stages in the new venture creation. One of the most common divisions of entrepreneurial stages is *prospecting, developing, and exploiting* (Bakker and Shepherd 2017; von Briel et al. 2018; Davidsson and Gruenhagen 2021). Prospecting involves exploration to detect signals of potential opportunities in a location or conceptual level, such as a sector of an industry, a set of information channels, a technological invention, or a latent market need. Developing involves advanced exploration of the signals detected through prospecting to generate further information about the feasibility and desirability of the potential opportunity for the focal actor. Exploiting involves building efficient, full-scale operations for products or services created by, or derived from an opportunity.

In addition to the process, several studies addressed how startups organize their activities. Shane (2003) explains that to profit from the discovery of an opportunity, an entrepreneur needs organizing, i.e., establishing the organization or market mechanism that she will use to exploit it. Organizing involves creating routines and structures that will be used to recombine resources into the product or service sold to customers and to create the entity that will undertake the recombination activity.

While the literature on NVC and DE provides a background on the nature of startups and how they differ from established companies, they mainly focus on the entrepreneurial process and organization. There are not many works that focus on how startups that have reached the exploiting stage organize their activities. As startups progress into the scale-up phase and begin to structure their operations, the domains of NVC and DE can derive advantages from the Business Process Management (BPM) discipline, the focal point of our work.

2.3.2 Startups capabilities and BPM

In recent years, several works in different domains have explored the startup context and capabilities. Table 1 shows these selected works, their motivation and findings. The previous works mainly focus on startup innovation and dynamic capabilities. El Hanchi and Kerzazi (2020) contribute to the evolving capability-based perspective on the innovation process by focusing on startups. They propose an innovation capability (IC) construct as higher-order DCs—sensing, seizing, and transforming. These dynamic capabilities (DCs) are sustained by a network of interconnected foundations: (a) Core Innovation Capabilities (ICs), encompassing technological IC, product IC, and business model IC; (b) Supporting ICs, acknowledging the significance of marketing capability, learning capability, and relational capability; and (c) Startup Entrepreneurial Capabilities, encompassing the founder's entrepreneurial and managerial skills, innovation funding capability, and resource management capability.

Goni and Van Looy (2022) focus on process innovation capability for less structured business processes. They propose a framework that comprises three pillars: people, process, and technology. They do not focus specifically on startups, although their proposed framework can be applicable to startups as startups tend to have less structured business processes.

de Oliveira Mota et al. (2021) highlighted the new product development as an important process to reduce startup time-to-market. They map the relationship between drivers and capabilities for reducing TTM in start-ups. The central factors in dealing with external uncertainties in a time-sensitive market are (1) management strategies that align with an organizational culture centered on time-based competition and (2) integration processes with other stakeholders in the supply chain, leveraging a robust capacity to absorb knowledge through effective communication channels and team experience.

Other works highlight the importance of dynamic capabilities in startups. de Santos and de Pádua (2023) argue that startups need favorable conditions to ensure their viability and internal capabilities for establishing a management system capable of addressing evolving product or service demands and strengthening the overall business. Thus, startups need to develop dynamic capabilities during their entrepreneurial process. They develop a BPM promotion framework that operationalizes the development of DCs in startups and then evaluate the implementation of the framework according to the principles of good BPM (Vom Brocke et al. 2014).

Griva et al. (2023) focus on growing startups at the early stage of the entrepreneurial journey, where process capabilities emerge. They argue that growth is different for startups compared to established firms. Startups have limited financial

Table 1 Selected literature related to capabilities and startups

Authors	Motivation	Contribution	Findings
El Hanchi and Kerzazi (2020)	Innovation capability (IC) lacks clear delineation as a distinct construct. The subject of startup IC is implicitly addressed within innovation management, entrepreneurship, or literature on small businesses	Suggests an IC construct as a higher order Dynamic Capabilities (DC) and three layers of foundations core IC, supporting IC, and startup entrepreneurial capabilities	The IC construct consists of capabilities for sensing the environment for opportunities and knowledge, seizing opportunities through innovation development and commercialization, and transforming founders' individual capabilities into organizational assets through effectuation
Goni and van Looy (2022)	Today's reality becomes increasingly complicated and knowledge-intensive, leading to more organizational needs for supporting less-structured business processes (LSBP)	Build process innovation capability (PIC) is becoming increasingly important to keep pace with changing business environments and emerging technological opportunities	Framework of Process innovation capabilities for Less-structured business processes, which comprise three pillars: people, process, and technology. The framework can help organizations to better utilize their resources to gain PIC through internal and/or external network relations i.e. help organizations to configure their PIC by profiting from the underlying theory-capacity basis
de Oliveira Mota et al. (2021)	No research aims to explore the reciprocal impact between capabilities and factors influencing the reduction of time-to-market (TTM), especially within the context of startup companies	Identify the map of the relationship between the drivers and capabilities for reducing TTM in start-ups	Five drivers led start-ups to attain nineteen capabilities to reduce their TTM. The capabilities present synergy and can be organized in a structural model that shows the hierarchy among the capabilities, indicating which one should be developed with priority

Table 1 (continued)

Authors	Motivation	Contribution	Findings
de Santos and de Pádua (2023)	Startups need to develop dynamic capabilities	Develop a BPM framework that operationalizes the development of DCs in start-ups	The BPM promotion framework for start-ups is divided into four steps: “frame BPM,” “understand BPM,” “enable BPM,” and “continue BPM”. Nine principles constitute the promotion of BPM: (1) context consideration principle, (2) holistic, (3) simplicity, (4) involvement, (5) empowerment, (6) shared understanding, (7) purpose, (8) technological appropriation and (9) continuity
Griva et al. (2023)	Growth is significantly different for start-ups as compared with established firms	Contribute to the definition of growth by examining fewer tangible dimensions Identify characteristics of early-stage digital start-ups experiencing growth	Growing digital start-ups exhibit a set of characteristics that include an agile culture combining clan with adhocracy; the ability to nurture their absorptive, innovative, and adaptive capabilities effectively; and human capital with adequate entrepreneurial skills, emotional attachment to and fitness with the start-up
Murmann and Vogt (2023)	New start-ups or diversifying entrants could develop products and services that potentially take sales away from established companies if the established companies fail to produce these products and services themselves	Develop a framework to assess dynamic competition that integrates ordinary and dynamic capabilities into the analysis by systematically comparing incumbents, start-ups, and diversifying entrants	Illustrate the framework with a case study of electric vehicles and aim to show how crucial such comparative analyses are for making well-founded predictions about the likelihood that incumbents will be able to maintain their leadership positions

performance, and their growth cannot be accurately measured by metrics such as customers, revenues, profits, and turnover at the early stages of the new venture creation process or before the start-up has reached sustainability. Digital startups that experience growth typically display a set of characteristics, such as an agile culture that blends clan and adhocracy, the effective cultivation of absorptive, innovative, and adaptive capabilities, and a human capital equipped with sufficient entrepreneurial skills, emotional commitment, and alignment with the startup's objectives.

Murmann and Vogt (2023) assert that established companies and startups must cultivate fresh capabilities in response to profound technological shifts. From the perspective of startups, the development of capabilities is usually seen not only through the lens of dynamic capabilities (DCs) but also within the context of cultivating ordinary capabilities (OCs). The challenges related to a lack of skills and scaling them up seem comparable to those encountered by established entities in acquiring new capabilities. The key distinction is that startups are not burdened by existing routines and mindsets when establishing new capabilities. Startups are not hindered by legacy issues that often obstruct the learning of new skills in established companies. However, scaling up new capabilities involves significant changes, as evidenced by research on firm growth. Unlike startups, established firms are constrained by existing routines and mindsets.

Previous studies have addressed different aspects of capabilities for startups from different perspectives. El Hanchi and Kerzazi (2020) acknowledge the entrepreneurial character of startups and underscore the significance of the entrepreneur's capabilities and behaviors in this context, but they do not specifically address process management capability. de Santos and de Pádua (2023) establish a framework to encourage the adoption of Business Process Management (BPM) in startups, yet their emphasis does not delve into how startups manage their activities. de Oliveira Mota et al. (2021) mention the interrelation among capabilities: strategies, culture, team/people, and method, but the IT and structure capabilities of startups are not specifically addressed. The capabilities highlighted by Griva et al. (2023) are culture, method, and people, but they do not specifically address strategy, governance and how IT can support other capabilities. Murmann and Vogt (2023) mainly focus on capabilities change and its impact on the culture and people's capabilities. It is essential to investigate the combined functioning of process management capabilities comprehensively to explain how startups holistically handle their processes. This is the central focus of our research.

3 Method

3.1 Design and sampling

This study follows an inductive approach (Sarker et al. 2018) that combines qualitative research with Grounded Theory Method (GTM) and emphasizes the meaning of observed data (Creswell 2014). We aim to investigate process management for digital startups in the scale-up phase of their entrepreneurial journey. Digital startups are chosen purposively by first compiling a list of potential startups and sending them a

proposal for participating in our research. After several iterations of discussion, two startups in the logistics sector agreed to participate.

Logistics startups focus on logistics innovation or breakthroughs for existing logistics service providers (LSPs) or their customers. Therefore, the value-creation process in this startup is closely related to the logistics activities of LSPs (Cichosz 2021). Startups in this sector develop various technology-based solutions that connect shippers with warehouses and transporters. Apart from that, startups in this sector also provide several types of services, such as multimodal transportation services, freight forwarding, shipment tracking and inventory management.

3.2 Case description

Table 2 summarizes the information about the two startups involved in this study. Lisy was founded in 2019 as a B2B logistics digital platform provider. Rise was founded in 2018 and is also a B2B logistics digital ecosystem provider.

3.2.1 Lisy

Lisy is a digital logistics platform provider that connects orders from shippers with a fleet of verified truckers throughout Indonesia. Through their platform, shippers can find trucks according to their needs and can check the location of trucks in real time. Founded in 2019, Lisy has four main services: trucking, intermodal, trailer and customs clearance, and logistics management. Lisy services have been trusted by hundreds of well-known shippers, including Unilever, P&G, Midea, Grab, and many more. In addition, Lisy also collaborates with the National Logistic Ecosystem (NLE), a logistics ecosystem that orchestrates relationships between logistics aggregator platforms throughout Indonesia. Lisy secured series A funding from Monk's Hill Ventures on March 20, 2020. Previously, Lisy also received seed funds on August 15, 2019.

3.2.2 Rise

Like Lisy, Rise provides a mobile and desktop application-based logistics service platform or transportation management system that connects shippers, transporters, and truck drivers through deliveries that can be monitored directly with detailed reporting. The startup, founded in 2018, has received three funding rounds. The last funding was series A, obtained on May 6, 2019. The services offered by Rise are divided into services for shippers and transporters. For shippers, Rise offers convenience in meeting trucking needs through applications supported by reporting systems, tracking, and API features. Meanwhile, for transporters, Rise can help optimize the trucks in delivering products to customers equipped with a profit reporting system.

Table 2 Case overview

Case code name	Lisy	Rise
Founding year	2019	2018
Founding context	Improve logistics delivery efficiency	Digitizing the logistics industry
Scope	B2B logistics digital platform	B2B logistic digital platform
Description	B2B logistics digital platform that connects shippers with trucks from verified carriers	Digital logistic supply and management platform with a B2B model
Industry	Digital logistics, supply chain	Digital logistics, supply chain
Digital market offering	A platform with an algorithm to aggregate supply and demand for trucking and provides shippers with trucks powered by real-time tracking and digital Proof of Deliveries (PODs)	A cloud-based platform that allows shippers, transporters, and drivers to process and save on freight costs through logistics planning software
Stage	Exploiting	Exploiting

3.3 Data collection and analysis

In conducting the data collection and analysis, we follow the GTM guidelines provided by Fernández (2003). The first step is entering the field. Although this research is motivated by previous research on BPM, we aim to uncover how digital startups manage their processes. Thus, the interview protocol and questions are designed broadly based on the work of Rosemann and vom Brocke (2015), who proposed strategic alignment, governance, methods, IT, people and culture as core elements of BPM. We also combined them with van Looy's (2019) BPM capability areas, namely life cycle, management, culture and structure. The guiding questions can be found in Appendix A.

Since most questions were open-ended, the original guideline served primarily as an orientation mechanism (Mueller et al. 2019). Follow-up questions were asked flexibly on the informant's responses. The selection of informants in this study used a purposeful technique because the informants were not intended to represent the population but rather to represent information (Creswell 2014). There are four informants from the two startups, as described in Table 3. Data are collected through in-depth interviews in Bahasa Indonesia between August 2021 and May 2022. All interview results are recorded and transcribed. The results are translated for the purpose of this paper.

Then, we conduct theoretical sampling, which is a data collection process that continues until the very end of the research. Data analysis with GTM can follow different processes of coding. In this research, we follow the Glaserian approach (Fernández 2003). We took the first slice of data from the interview results regarding how they develop their processes. The interview results are read line-by-line and labeled, called open coding. We use computer-assisted qualitative data analysis software (CAQDAS), NVivo 12, to support the coding. During this process, we were primarily concerned with understanding the concept, and what is going on rather than the words, as suggested by Fernandez (2003). Examples of open coding that we obtained regarding how Lisy develops their process are summarized in Table 4.

As we conducted open coding, we also started making categories and their properties. According to GTM guidelines, researchers must ensure constant comparison (Urquhart et al. 2010) i.e., constantly comparing instances of data labeled as a particular category with other instances of data in the same category. For example, as we code "solve problems", we also identify "create value". As we added more slice of data, we discovered another code, "understand the problem" by "root cause

Table 3 Information about informants

Informant	Startup	Position	Informant code
RB	Lisy	Co-founder and Chief Technology Officer	LS1
KR	Lisy	Operations Manager	LS2
RT	Rise	Head of Operations	RE1
FR	Rise	Customer Experience Manager	RE2

Table 4 Example of open coding

Interview	Excerpt	Code
Lisy-1	The process arose to solve certain problems	Solve problems
	The process is not the goal, it is the way to get it done	
	Instead of process-wise, we see the organization as more analogous to the problem being solved	
	Maybe it can be called a process, but maybe I look more at how the objectives of the smaller organizations are derived, and for that objective, what are the steps for solving this problem	
	Yes, because every process must have a pilot. After the pilot, go live, run smoothly, then monitor. Because we see that process solves problems. We see whether the problem is still there or not	
	If it does not solve certain problems, it creates some value in the process. for example, value in a sense of innovation	
	What is the problem/value we want to create, let's start from there	
	So we always look there first, what innovation or problem we want to solve first	
	When we face problems/values, we also learn what the root cause is. Because often, when we see a problem, it's not good to jump straight to the solution. We must know what is causing this problem	
	So, the process or solution to the problem often appears after we understand what the root cause of the problem is	
	So, we always start from there: process problem, then we understand the solution, understand the root cause first, then we make hypotheses	Create value
		Understand the problem, root cause

analysis,” which relate to “solve problems”. This is categorized as process origin. Examples of the open coding and categorization of interview results from Lisy can be seen in Table 5.

While conducting open coding, we also write memos where we write the ideas about codes and their relationships. Writing a memo allowed us to understand the categories and note further information that needed to be collected. The next iteration of interviews is needed to complete the information gap. This is also our way to follow the guideline of GTM, *theoretical sampling*. The interviews and open coding stages are conducted several times. As we start with Lisy, three rounds of interviews are conducted with additional validation interviews, after which no new information is obtained. With our experienced with Lisy, in Rise’s case, we conducted three rounds of interviews when no new information was obtained. Table 8 in Appendix B summarizes the interviews, duration, and scope. The interviews with informants lasted 24–60 min. Table 9 in Appendix B summarizes the information obtained in each interview, an example of the memo, and findings from every iteration.

As codes and memos accumulate, we try to increase the level of abstraction and relate categories to each other through iterative conceptualization. The code and categories from both cases can be found in Appendix C. We try to find relationships between the categories, which in GTM is called Theoretical Coding. We conceptualized the interrelation of substantive codes by generating hypotheses for integration into a theory.

Theoretical coding results from each startup, Lisy and Rise, are compared to find patterns across the two case studies, i.e., selective coding. The aim is to theorize the management of business processes run by logistics startups by abstracting the findings from the two startups. By cross-case analysis, we also follow the guidelines of GTM, i.e., scaling up, to ensure the generalizability of the theory. Table 6 shows categories that were found in both startups. By comparing the ten main categories found in the two case studies and slightly adjusting the ten parts of these categories, we found seven main categories: (1) industry and stakeholders, (2) digital offerings drive processes, (3) organic structure, (4) process management, evaluation and improvement, (5) performance measurement, (6) employee training and (7) culture.

As shown in Appendix B, after four iterations for Lisy and three iterations for Rise, we could account for the main concern of the research, and further sampling failed to add significant value to the study (i.e., new categories or properties), i.e., Theoretical Saturation. Finally, we refer to previous literature on BPM and startups to achieve theoretical integration. We found four key themes: the competitive nature of startup, adaptive process experimentation, agile people, and culture, and organic structure for our theoretical model.

4 Findings

4.1 How logistics digital startups manage their activities

Following the data analysis, we learn how logistics digital startups Lisy and Rise manage their activities. As startups operating in the logistics sector, Lisy and Rise

Table 5 Open coding and categories of Lisy interview 1

Code	Subcategories	Category
Solve problem		Process origin
Create value		
Understand problem		
Root-cause		Process documentation
<i>Business as usual is documented</i>	–	
SOPs evolve following the issues addressed		
It used to be a process centered on someone		Process evolution
In the past the process was not documented	–	
Now the process depends on the function		
Processes ensured to be documented		
<i>Managed service level</i>		
Many processes have not yet been formally arranged		
Technology automates processes		The role of technology
Technology improves process efficiency	–	
<i>Reward</i> for milestone achievement (ad hoc)		
<i>Reward</i> for performance achievements (routine)		<i>Reward</i>
The sales department has a <i>reward system</i> from the beginning		
<i>Rewards</i> for playing together early in development	–	
The <i>engineering</i> department has a <i>reward system</i> in the first 6 months		

Table 5 (continued)

Code	Subcategories	Category
Customer query search (<i>sales</i>)	Sales of products to customers (<i>commercial</i>)	The three main functions of Lisy
Supplier/transporter search (<i>transporter management</i>)		
Customer needs fulfillment (operational)		
Problem search and solution (<i>product</i>)	<i>Product</i>	
Platform developer (<i>engineering</i>)	Secretarial	
<i>Finance</i>		
<i>People operations</i>		
<i>Validated learning</i>	–	<i>Build-measure-learn</i>
The organization is still too small	2019 has not implemented OKRs	OKR
Objective Key Results (OKRs) are derived from the organization's vision and mission	OKRs for each team member	
<i>Performance development and discussion</i> every semester based on OKRs	–	

Table 6 Similar categories in both startups

No	Main category	Similar code
1.	Partner support	Transporter Shipper Investor
2.	Process origin	Processes exist to solve customer problems or needs
3.	Organic organizational structure	Division Sales or business development Operations: control tower, proof of delivery, field operations Transporter management Operational excellence IT or tech: product, engineer Finance HR Leadership level inconsistency No dedicated function managing the process Organizational structure evolves following the need for job specialization
4.	Process management	Cross-departmental collaboration Process documentation in the form of SOPs containing flowcharts Product management The process continues to evolve so that the process documentation follows
5.	Ad hoc process evaluation	Reactive or as needed
6.	Process improvement	Process improvement depends on the organization's knowledge of the importance of process improvement and how to do it
7.	Performance measurement	Individual targets achievement (OKR or KPI)
8.	Employee training	Basic and soft skill training
9.	Agile culture	Open communication Open to team ideas (freedom of change) Agile, fast-paced working environment Open to customer input
10.	Rewards	Rewards based on individual performance achievements

create and provide digital products for their customers. Lisy is a business-to-business (B2B) logistics startup with core services in the trucking field that brings innovation to the ordering process, truck unit search, truck tracking, and assistance for customers supported by technology. Lisy also expanded its services into the import–export field in response to customer needs. Rise has two main services. The first is software as a service (SaaS), where Rise only sells applications to customers. The second is the trucking marketplace that connects cargo owners with truck vendors throughout Indonesia.

4.1.1 Industry and stakeholders

The nature of the industry and digital products influence how both startups manage their activities. As they are involved in B2B, instead of consumer products, both startups serve and work closely with their customers and partners. Considering that both have trucking services where they do not own the trucks, the transporters, as the truck owners, are the key customers for their business. Besides that, the logistics business cannot operate without companies or cargo owners that need to send the goods (shippers). In this case, the shippers also become customers for the logistics startup.

The next partners commonly found in most startups are investors. Investors who inject funds to invest in startups expect a profitable return on capital in a relatively short period of time. Investors also provide support in other forms, such as connections and mentoring facilities. In return, startups are indirectly challenged to develop business ideas quickly in order to return the capital obtained from the investors. This is one of the driving factors for rapid development in the logistics startup environment.

4.1.2 Digital offerings drive processes

As pointed out by Lehmann & Recker (2022), digital startups provide digital offerings that are inherently malleable, so the digital market offerings are ever incomplete and in the making. Lehmann & Recker (2022) explains that such digital ventures continue product development post-launch.

As explained by the CTO of Lisy:

“Our business is not a new business. What we are doing now is actually already being done by other companies conventionally. So, the problem–solution fit has actually been around for a long time, and it was carried out by companies that we consider conventional, such as DHL, Kuehne+Nagel, and so on. But, we see several problems: scalability, reliability, etc. That’s where our product, enabled by technology, can enter...Today, when you ask for a certain kind of truck, we will start looking for trucks and matching them. All of this is supported by the application, and it is the technology that allows us, with this limited number of people to be able to serve several orders a month, which is what conventional companies can’t dream of. But if you ask whether this is a steady state, I don’t think so”—LS1.

Processes are developed in accordance with customer needs. Both startups in the logistics sector are driven by customer problems and needs, which influence their strategies to achieve startup goals. Although both startups already have a product that is accepted by the market and gives them profit, they consider processes as a means to solve customer problems.

In Lisy’s case, the process is structured by first understanding the root of the problem faced by the customer. Processes can also be designed based on the value or innovation they want to provide the customer. As described by the following excerpts.

"The process arose to solve certain problems." (RE1)

"So we always start from there: the problem then we ... understand the root cause first, then we make hypotheses. If it does not solve certain problems, it creates some value in the process, for example, value in a sense of innovation."—(LS1)

"What I want is for Lisy to be able to solve this specific logistical problem. When the motivation is solving problems or solving pain from your customers, you will have a reason to keep going as long as you know that your customers are still having this pain. And that is such a strong motivation to keep going"—(LS1)

The processes in Rise are determined based on activities commonly found in the logistics sector. Customer needs drive Rise's innovation projects. This happens if the customer needs certain features or products that will then be pursued by Rise. In addition, innovation can also be driven by the internal needs of the organization in developing its business. Therefore, processes can also arise from the organization's need to realize such innovations. However, Rise as a startup does not have a comprehensive business process architecture. Business processes are structured reactively.

"Do we innovate following the customer? Following customer needs, but not following customer requests, because requests and needs can be the same, or they can be different. [...] But in the end, we strive to provide good service to customers"—(RE2)

4.1.3 Organic structure

The organizational structure of the two logistics startups is adapted to practices in the logistics sector. There is a division that takes care of customer interests, which is usually called sales or business development. There is also a division that ensures the delivery process is carried out safely and on time, or what is called operations. Usually, operations consist of several other teams, such as the control tower, which monitors the movement of goods 7×24 h; proof of delivery, which handles waybill documents; and field operations, which monitors the loading process at the location of the cargo owner (shipper). Apart from that, a division is tasked with acquiring transporters and managing transporter performance. Usually, this division is called transporter management. Furthermore, there is an operational excellence division whose role is to ensure the smooth running of business operations.

Considering that both startups are digital startups, there is a tech division that is responsible for the digital product offerings. This tech division consists of a product team, which is responsible for finding problems and designing product solutions, and an engineering team, which technically develops applications. To support the organization's main functions, startups have a finance division, which handles the organization's finances, and a human resources or people operations division to manage human resources. However, both startups do not have a specific function that focuses on managing processes.

Despite divisions commonly found in logistics startups, the level of leadership in each division is not the same. There are divisions led by heads, and some are led

by managers. In addition, the organizational structure continues to evolve to suit the needs and complexity of work.

“Not every division has a head, and not every division has a VP. Some are VPs, some are heads, some are without VPs, and some are VPs without heads. Some don’t have a manager”—RE2

In addition, the organizational structure continues to evolve according to the needs and complexity of work. The organizational structure of logistics startups tends to be fluid and can change easily in a short time.

“So our department is dynamic. We change divisions/departments in a year, maybe three times, because we adjust according to the needs. It’s also possible that several departments that were not needed before then appear now, ... because our business needs it now,... what was previously unnecessary now seems necessary. So we create it. We often restructure”—RE2

4.1.4 Process management, evaluation and improvement

The process at a logistics startup involves cross-division collaboration, although there is still work that each division must do. Logistics startups do not implement a way of working that divides divisions because they have the principle that the most important thing is that organizational goals can be achieved. The processes they carry out are documented in the form of a flowchart. This documentation is not as neat and complex as the process documentation that traditional companies have. Since logistics startups can quickly absorb and make changes, documentation of indirect processes also continues to change following developments in the processes being carried out.

Because the main products of the two logistics startups are digital, product management is inseparable from process management. Product management is considered more structured than other existing processes. This condition is very normal because the digital offerings are the core of their business. So much attention is focused on product management.

“Because we always innovate... So we must find ways to release new products because we want to sell many things to our customers. Releasing new products needs to be fast because it takes a lot of trial and error... we don’t even know if this product will be successful or not”—RE2

Lisy applies the RACI matrix to determine who is responsible for a process. Cross-departmental collaboration is also commonly done, especially if there is a unique customer request. Lisy already has process documentation in the form of standard operating procedures (SOPs). This SOP consists of a sequence of activities and a flowchart. Most stable business activities, such as payroll systems, are well documented. There are also SOPs owned per department, for example, SOPs specifically for operations departments. However, given the dynamic startup conditions, documentation of this process continues to evolve following the context of the problem that is addressed according to customer needs.

In carrying out their activities, each team at Rise certainly has their own work. However, they are also often involved in projects that require collaboration between teams. Given that it has been certified ISO 9001, Rise has process documentation in flowcharts. However, the documentation was not reviewed further. In addition, Rise also has project and order management, each of which determines the division that runs it.

“Sometimes some of the process must be done quickly. Suppose we want to roll up a certain division or a certain activity. In the end, the analysis of the business process was not carried out in a documented way. Maybe it was just an interdivisional discussion. We will do A, B, C D, okay? It’s not done comprehensively because, in the end, it needs to be faster—RE2

Lisy also swiftly evaluates the processes. When they find inefficiency in delivering services to customers, then improvements can be made right then and there.

“We can do that [process improvement] real quick, yeah. For example, at our weekly meeting, we figure out something that is lacking in our process because our customer feels that they are having trouble with it. It can happen in a very short period of time. For example, within a week, we figured out that our existing process was inefficient in satisfying our customers, so we could change it. It’s something that happens daily” – LS1

In addition, Lisy also has one pillar, namely the customer-obsessed pillar. Through this pillar, Lisy can assess whether customer problems have been resolved with the services provided. One measure is using a net promoter score (NPS). Rise does not evaluate the process, but only measures the end result of a particular process. If they find a problem in their business, the evaluation of the new process is carried out reactively and as needed only.

In Rise, process improvements depend heavily on the awareness and knowledge of the people within them. This is stated by the Customer Experience Manager of Rise as follows:

“When things do not meet our customers, we should reflect. Several divisions do this... they focus on it, take measurements, and continue to make improvements. But, some don’t go there even though it’s a problem. Because the scope of customer experience is quite wide. We will also help them with matrices, etc. Usually, we start with the problem raised by the customer. For example, a long payment process. From there, we will get insight, will try to give it to the finance team, and discuss ways to improve... But it is not carried out continuously. Maybe because they are unaware. Or they are aware of the needs, but they don’t have the time and resources. There are several divisions that don’t do that, even though it’s important. That’s quite a challenge for us.”—RE2

Lisy uses several methods to determine follow-up process improvement, including root cause analysis, sprint retrospective, and postmortem analysis. Root cause analysis is commonly used by engineering teams in the event of a system failure. Sprint retrospectives are used to determine improvement initiatives in future sprints.

Meanwhile, postmortem analysis is used at the executive level related to large projects and external parties. From these methods, new initiatives will be generated to improve the implementation of the process. Unlike Lisy, process improvement is not carried out continuously in Rise. Not all divisions make improvements to the process. If any improvements are made, they may still be local in their respective divisions, causing overlap with other processes.

4.1.5 Performance measurement

Every Lisy employee has their own targets that must be achieved in a certain measure. Objective key results (OKRs) measure progress towards achieving these targets. This OKR is derived from the organization's strategic objectives to the staff level, so that everyone has an OKR, which will be assessed every semester through the performance development and discussion (PD&D) agenda. Employee performance measurement carried out by Rise is by measuring the achievement of KPIs from each member each year. This assessment refers to the performance plan index (PPI) prepared at the beginning of the year from discussions between employees and their leaders at the head level. This plan is then assessed through an index performance review.

4.1.6 Employee training

Lisy provides training for its employees, including basic training, soft skills training, and specific training needed by certain divisions. Basic training that is usually carried out regularly covers general skills, for example, operating systems used daily, such as Google Workspace. Training is also related to certain soft skills, such as problem-solving. In addition, Lisy also facilitates its team to attend special training to support work in a division, for example, related to export-import.

As an effort to support employee competence, Rise provides several training. The main training provided is related to product knowledge to ensure that employees master and understand the products they have. There is also training that supports soft skills. However, this soft skills training is usually initiated by each division which can be followed online.

4.1.7 Culture

Customer problems and needs are the main driving factors behind all activities and innovations carried out by Lisy. To capture customer problems and needs, Lisy always instills an open mentality to all inputs that can improve their services. Lisy is also open to ideas or proposals for process improvements from its team members, considering that they are the ones who carry out these activities daily and understand the conditions of implementation. Lisy employees can work from anywhere, not having to come to the office every day. The flow of communication between the employee and leadership levels is also very open. Employees can deal directly with the CEO. In addition, Lisy has a very supportive environment for its employees to continue learning. Moreover, the demands of competitors and customer needs drive

Lisy to constantly look for the best way to provide its best solutions and services. Lisy has the tendency to recruit potential employees who, within 3–6 months, can be promoted to higher posts.

Rise has a work culture that is almost similar to most startups. Their communication style is very flexible, and employees at various levels can communicate directly and informally. Employees can communicate directly with C-levels. At the very least, employees must inform their interests to the manager or leader above them. Rise also gives freedom to his entire team to initiate change. So, they have a work environment that changes very quickly.

Some rewards given to Lisy employees are routine, and some are ad hoc. Routine rewards are given based on the achievement of each employee's OKR, which is assessed through the performance development and discussion (PD&D) agenda. Meanwhile, ad hoc rewards are given if there is a certain milestone achievement, for example, new funding is obtained. The rewards given by Rise to its employees are based on performance appraisal. In addition, Rise also provides rewards for employees involved in certain projects.

4.2 Theoretical model: startups process management

Based on the findings from the case studies, we develop a theoretical model of startup process management shown in Fig. 1. The main notion of the model is the characteristics of startups that differentiate them from established organizations and influence their process management.

4.2.1 Competitive nature of startup

As shown in Fig. 1, the competitive nature of startups plays an important role in how they manage their activities. Startups operating in a B2B sector, as opposed to consumer goods, consider the customers as close partners. Although both startups already have a product that is accepted by the customers and gives them profit, they

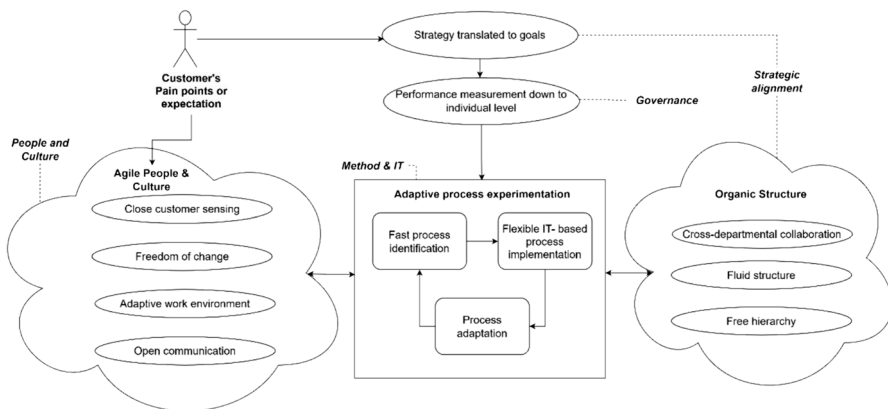


Fig. 1 Process management in digital startups

consider processes as a means to solve customer problems or provide value to the customers.

The fact that they are digital startups where IT not only supports the process but also becomes the output and business model also plays an important role in driving the startup processes. Digital products are malleable to accommodate customer problems and needs. These drive startups to continuously sense the customer pain points or expectations, and improve their digital offerings. The competitive nature made the startups place a strong emphasis on speed and agility. The values are translated into how they define their goals and performance measurement. The startup's goals are translated into several key results as targets for each division and down to each employee. Our findings that one of the startups uses OKR (de Oliveira Mota et al. 2021) as a goal-setting framework that aims to align the organization and its teams toward common objectives. The other startup uses the Index Performance Plan at the individual employee level. These highlight that startup divide their goals down to the individual employees and carefully measure them as the basis for promotion and reward.

The achievement of the startup's strategy and goal is influenced by three components: the adaptive process experimentation, agile people and culture, and organic structure. Digital startups that already reached the scale-up phase already have typical processes in place, including core, support and management processes (Harmon 2015). For example, in the two logistics digital startups, the core processes include product and operation. The support activities include back office functions such as finance and human resource management, while the management process includes defining strategic goals.

The difference with traditional organizations is digital startups focus heavily on product development of their digital offerings. Hence, the process improvement cycle is intertwined with digital product development. Digital startup continuously improves their product and conduct what we call this adaptive process experimentation to support product development. The theoretical model also shows supporting capabilities in which the startup operates: organic structure and agile people and culture. These capabilities (structure, people and culture) enable the process management of startups.

4.2.2 Adaptive process experimentation

Adaptive process experimentation means startups conduct repeated product release experimentation and validated learning. The experimental cycle of product/process experimentation occurs continuously consisting of fast process identification, flexible IT-based process implementation, and process adaptation. Our conceptualization aligns with El Hanchi and Kerzazi's (2020) innovation capabilities (IC constructs) as higher-order Dynamic capabilities composed of three DCs—sensing, seizing, and transforming—and three layers of foundations—core IC, supporting IC, and startup entrepreneurial capabilities.

Putting the customer at the center of all business activities is known as customer-centricity (Frank et al. 2020). Startups attempt to sense the customer's problems and use this as a trigger for product improvement. They will first understand the root of

the problem faced by the customer and then propose potential solutions. In other situations, startups recognize innovation opportunities that can provide customer value. This “close customer sensing” must be done quickly as startups need to seize the opportunities. Thus, we termed this as fast process identification, which requires individual capability to recognize customer issues and organizational capability to sense potential innovation opportunities. El Hanchi and Kerzazi (2020) conceptualize these capabilities as core ICs, particularly product and process ICs. Product and process IC entails all capabilities that facilitate identifying product/process opportunities, idea generation, and seizing opportunities through the design, conceptualization, development, and implementation of new products and processes. It also involves transforming product strategy to effectively respond to and shape new market conditions.

Once the root-cause of customer problems is identified, startups must design and implement the solution in the product offerings. This fits well with Badakhshan et al. (2020) description of agile capability, i.e., “continuously codifying and integrating requirements, and understanding, documenting, designing, analyzing, simulating, executing, and monitoring business processes”. Startups require flexible methods and tools to rapidly implement the improvement in their digital offerings as proposed by El Hanchi and Kerzazi (2020) Technology IC. Technology IC encompasses all the necessary capabilities for sensing technology, engaging in research and development, seizing technological opportunities, harnessing new technologies, and effectively managing technologies to adapt to changes and influence emerging technological scenarios.

Once the new process is implemented, startups adapt it to the new solutions. The improvement process is not defined as in the classical BPM lifecycle but reactively conducted based on customer value identification or as a result of process experimentation. The process improvement, in return, triggers further alignment of products to market. This process adaptation relates closely to agile people and culture and organic structure. Without these supporting capabilities, the change will be difficult to achieve.

4.2.3 Agile people and culture

As suggested by Badakhshan et al. (2020), agility requires the ability of organizations to be continuously ready to create change and embrace change. Agile culture is argued as an important characteristic of startups. We further theorize that such capability needs to be nurtured at the individual and organizational levels. At the individual level, Baiyere et al. (2020) suggest that to buffer against the tensions in transformation, organizations need mindful actors.

People in the startups, particularly those involved directly with the customers, need to nurture their sensing skills. This aligns with the previous work of Schmiedel et al. (2014) and Schmiedel et al. (2020) that propose customer orientation as one of the four values constituting BPM culture. To ensure that innovation is nurtured, startups give freedom of change to their employee. The startup work environment must adapt to continuous change driven by product/process innovation.

Finally, open communication is required to ensure that everyone involved in the changes is kept informed. Open communication also means that there is less bureaucracy, so information can be transferred and decisions can be made quickly.

4.2.4 Organic structure

Another salient factor for startup process management is organic structure. In traditional organizations, the structure is typically rigid and can be considered fixed for a certain period. This allows for a more efficient workplace where everybody knows what they are doing and a lower cost for coordination and supervision. However, it also restricts the organization's ability and speed to change.

Our finding suggests that startups can retain agility because their organizational structure can easily change to accommodate their needs. People are assigned to projects where they implement cross-functional collaboration. The structure is fluid and can be easily changed to accommodate startups' needs. Finally, it does not have a rigid hierarchy. The managerial level depends on the scope and span of the activities.

The organic structure and agile culture influence one another. A structure that is organic requires people that are flexible and ready for change. On the other hand, the nature of organizations that always change also creates an agile culture.

5 Discussion

After describing the findings from two logistics digital startups and theorizing how digital startups in the scale-up phase organize their activities, we will discuss the implications of our findings on the BPM discipline and NVC/DE domains.

5.1 BPM for startups

Startups provide an interesting context for the BPM discipline. The nature of startups challenges the classical BPM discipline in several ways. Table 7 summarizes the comparison of BPM capabilities in traditional environments compared to the startup environments, which opens up opportunities for further studies.

5.1.1 Strategic alignment

Startups focus on agility and speed rather than efficiency and compliance, so they favor agile BPM more than the classical BPM. Furthermore, their strong focus on customers also aligns with customer-centric process management (Frank et al. 2020). Although many traditional organizations also emphasize customer centricity, in startups, customer needs, and values drive the process. The question is: how do different strategies influence startups at the scale-up stage success? There is also a need to develop a tool to help startups assess how well they are managing their processes. The maturity model developed in the BPM literature needs to be adapted to the context of startups.

Table 7 BPM capabilities in traditional versus startup environments and opportunities for further study

BPM Capability		Traditional	Startup	Opportunities for further study
Strategic alignment	Value: compliance, efficiency, quality	Value: speed and agility		How do different strategies influence startups at the scale-up stage success?
	Strategy is driven by vision and mission	Strategy translated into individual goals or performance		What is the best “process management maturity model” for startups?
Governance	KPI-based	Individual-based performance measurement		How to measure success for startups at the scale-up stage?
				What are the governance characteristics of startups? What is the impact of process agility on process compliance in startups?
Structure	Fixed organizational structure	Organic organizational structure		What is the impact of organic organizational structure on the individual, team, and startup level What other type of structure is adopted by other types of startups?
Method	Compliance based BPM	Agile BPM		Method to visualize startup process architecture Modeling method for startup processes
				Method that is agile but ensures compliance How can customer-centric design be applied in a startup environment?
IT	Support	Means to create value		What type of BPMS work for startup? How can IT integrate core and support processes in startups?
People	Procedural	Agency Mindful		What individual capabilities are needed for startup process management? What individual characteristics fit the startup environment?
				How do individual employees adjust to the startup's organic structure?

Table 7 (continued)

BPM Capability	Traditional	Startup	Opportunities for further study
Culture	Traditional	Agile	Do the values of excellence and responsibility apply in the context of startups? How do different startup cultures contribute to startup success? How do startups change old habits and routines as they scale up?

5.1.2 Governance

Governance refers to the capability to establish relevant and transparent accountability and decision-making processes to align rewards and guide actions (Rosemann and Vom Brocke 2015). The question is whether startups in the scale-up phase can be considered as established firms. How to measure success for startups at the scale-up stage and what are the governance characteristics of startups? Startups focus on agility, but what is the impact of process agility on process compliance in startups?

5.1.3 Structure

Unlike other capabilities, the structure received less attention in the literature, which opened up opportunities for further studies. In our studies, we discovered a structure that is “organic”, i.e., easy to change, which raises two questions: what is the impact of organic organizational structure on the individual, team, and startup levels? Further study needs to be done to determine whether such organic structures are also found in other types of digital startups. What other types of structure (if any) are adopted by other types of startups? Another interesting question will be understanding how to balance organic structure with compliance and efficiency.

5.1.4 Method and IT

The nature of startups fits more with agile BPM than compliance-based BPM. Works on agile BPM seek to explain how agility and BPM can be integrated systematically through the three main components of agility: flexibility, leanness, and continuity (Badakhshan et al. 2020). Our findings add to this stream of literature by providing insights into how startups practice agile capability. The two startups reveal that they continuously improve their products and services to cater to customer needs. This aligns with the continuity taxonomy of agility proposed by (Badakhshan et al. 2020), which emphasizes the need for organizations to be continually ready and have minimal time and cost to prepare the component for use. Secondly, the startup cases also emphasize flexibility, which they continually seek ways to improve their product/service to customers leading to process change and improvement. The challenge is developing a method that is agile but ensures compliance.

Our results also aligned with the three logics of BPM in the context of digital transformation proposed by Baiyere et al. (2020). Our findings show that digital startups continuously experiment with their digital offerings. Therefore, instead of rigid processes, they are more suitable with light touch processes, which are structured to be modifiable rather than rigidly fixed. As highlighted by Mendling et al. (2020), in a dynamic world driven by digital innovation, the classical BPM assumption that considers process design as finding a solution to a given process-related problem has become obsolete. Instead, they suggest that problem understanding now drives solution understanding and vice versa. They proposed that fast feedback and short learning cycles are required to test which process design works best given the current business environment. They also add the needs to balance predefined structure with freedom for adaptation. This is also what we found in our cases as

the startups continuously try to test their product and processes to meet the requirements of the customers. An interesting opportunity for further study is to see how the customer-centric design proposed by (Frank et al. 2020) be applied in a startup environment.

We also found that in the digital startup process, process-related methods are intertwined with IT project capability. Startups use root-cause analysis together with scrum (also suggested by de Oliveira Mota et al. 2021), sprint retrospective, and post-mortem analysis. Startups do not have process architecture. But, they may benefit from a visualization of their process architecture. Further study can develop a method to visualize startup process architecture. Startups use simple flowcharts rather than more advanced process modeling tools. The question is: what is the best modeling method for a startup?

5.1.5 People

Another important capability is that continuous change in startups requires people who are always ready for change. So the third logic proposed by Baiyere et al. (2020) of mindful actors, i.e., giving the actors enacting a business process the agency to evaluate a business process and their corresponding actions based on the prevailing circumstances of the context, is crucial in the startup environment. Several questions that need to be investigated here are: what individual capabilities are needed for startup process management? What individual characteristics fit the startup environment? How do individual employees adjust to the startup's organic structure?

Further research can also investigate further regarding the mindful actors' logic. What capabilities need to be instilled? How to create such skills for employees? Also, how do organizations balance developing mindful actors while ensuring accountability?

5.1.6 Culture

In terms of culture, we found that customer orientation and teamwork out of the four values of BPM culture in the startups. It will be interesting to explore if the other two constructs of BPM culture, i.e., excellence and responsibility, can also apply in the context of startups. How do different startup cultures contribute to startup success? How do startups change old habits and routines as they scale up?

5.2 NVC and DE process management

So far, studies about startups on NVC, DE and other literature have focused on how NVC reaches the scale-up stage and what factors contribute to the success. Literature in these domains has focused on the entrepreneurial process instead of processes conducted to run the startups. There needs to be further studies on what happens after the startups reach the scale-up phase. Do startups behave as established firms after they scale up? Our study theorizes that they retain their focus on the customer, focus on speed and agility, and continuously experiment with their product

offerings. This highlights that the NVC and DE domains will benefit from the BPM discipline. Further studies can try to find different ways in which startups should manage their processes and the impact on startups' performance.

There are different types of IT-associated startups, including startups where IT facilitates making the startup's operations easier, mediates new ventures' operations, and becomes an outcome of entrepreneurial operations. Compared to digital startups where product and process management are intertwined, the role of IT is less ubiquitous in these three startups. The question is, how do these kind of startups manage their processes? What differentiates these startups in their process management from established firms?

5.3 Practical implications

Findings from this research also have implications for practitioners. Recent incidents involving startups mean that startups must find a way to maintain speed and agility while balancing them with other values such as compliance and trustworthiness (Rosemann 2019) and other values such as environmental, social, and governance (Rosemann et al. 2023). Furthermore, agility requires an agile culture and people. Startup managers must balance the pressure for agility on the individual, team, and organizational levels.

5.4 Limitations of the study

We presented results from two digital startups in the logistics sector, which limited our findings. The results are influenced by the characteristics of the logistics sector and digital startups. While we are able to reach a certain level of theoretical saturation from the two case studies, we acknowledge that the theory needs to be validated further. Further study can be done by replicating our approach on other cases within the same sector, the same stage of the entrepreneurial phase (scale-up), and cases in different sectors. This way, we can increase the generalizability of our theoretical model. In this study, we also took a snapshot view of startups in their current condition. Considering the dynamic nature of startups, it will be beneficial to see how the startups develop over time and how they change the way they manage their processes. We focus on process management from a BPM capability perspective. It provides us with a holistic overview of the capabilities and their inter-relations. But it also means we cannot go into detail on each capability. Finally, we present the results without considering the performance of the startups. Further study may develop a measure of success for scale-up startups and see if the process management capability relates to their success.

6 Conclusion

This paper aims to understand how startups in the scale-up phase manage their activities. Findings from two logistics digital startups allow us to identify important categories that play a role in how the startups manage their activities: industry and stakeholders, digital offerings, organic structure, process management, performance measurement, employee training and culture. We further theorize the competitive nature of startups, i.e., operating in a B2B sector, as opposed to consumer goods, and providing digital offerings for their partners, make startups customer-centric and focus on agility and speed. Digital offerings are malleable and ever-in-the-making. Although both startups have reached the scale-up phase, they continuously sense customer pain points and opportunities to create value and improve their digital offerings. Digital startups focus on the product development of their digital offerings as their core process. The process improvement cycle is intertwined with digital product development. Digital startups continuously improve their product and conduct what we call this adaptive process experimentation to support product development. The supporting capabilities in which the startup operates are an agile people and culture, and organic structure. Agile culture is reflected in close customer sensing, freedom of change, adaptive work environment and open communication. People need to be open and ready for change. The organic structure emphasizes cross-functional collaboration, fluidity to change and free hierarchy. These capabilities (structure, people and culture) enable the process management of startups.

Our paper contributes to extending the current BPM literature with new insights regarding how digital startups, as a different type of organization, have different characteristics that lead to a unique way of managing their processes. At the same time, we also contribute to the NVC and DE regarding how the scale-up startups manage their processes. We provide several opportunities for further study in the domain of BPM for startups.

Appendix A: Guiding questions for interviews

Code	Question
BG	How was this <i>startup</i> originally built? What is the drive/background, what ideas/solutions do you want to develop?
P1	How are startup activities and work developed?
P2	Are <i>the inputs</i> , activities, and <i>outputs</i> of all jobs identified and documented? For example, through SOPs
D	Is there an information technology (IT) system that supports <i>startup activities</i> ?
C1	What is the mechanism for evaluating the performance of startup activities and work?
A1	How is the achievement of performance targets (KPIs, etc.) of organizational activities followed up?
A2	How is innovation/improvement of activities and work carried out following existing developments?
PS1	How are activities and work related to organizational strategy?

Code	Question
PS2	How does organizational strategy translate into objectives used to manage activities and work?
ER1	How <i>do startups build relationships with stakeholders</i> (customers, suppliers, investors, and other partners)?
ER2	How <i>do startups</i> know and understand <i>customer expectations</i> ?
RR1	How are the tasks and responsibilities divided at <i>startup</i> ?
RR2	Is there a process management team dedicated to structuring, analyzing, and improving startup activities and processes?
ST1	What is the team's ability to manage activities and work?
ST2	Is there training for the team on a specific activity or job?
V2	How is the startup's commitment to change, <i>continuous process improvement</i> , and innovation?
AR1	What is the mechanism for hiring, developing, and rewarding <i>startup teams</i> ?
AR2	What is the basis for awarding?
TM1	Is there an executive team responsible for managing a process-oriented way of working? For example, <i>chief process officer</i>
TM2	How does <i>the startup's</i> executive team view manage activities and work?
OC	Does <i>the startup</i> have an organization chart that explains process-oriented roles? Are activities and jobs explicitly displayed in the org chart?
GB1	Are there teams/units that support innovation, change, and implementation of activities/work?
GB2	Is there a role/unit within the startup that is responsible for the overall performance of the organization's activities and work?

Appendix B: Interviews, interview findings and integrative memo

See the Tables 8 and 9.

Table 8 Data collection interviews

Startup	Date	Informants	Duration (minutes)	Scope of interviews
Lisy	25 January 2022	RB (LS1)	72	Exploring information related to the management of activities
	15 February 2022	RB (LS1)	68	Further exploration into structure, teamwork and process documentation
	14 March 2022	KR (LS2)	97	Exploring information related to organizational structure and culture
	13 April 2022	RB (LS1)	46	Validation related to the findings from the previous interviews
Rise	13 April 2022	RT (RE1)	24	Exploring information related to the management of activities
	10 May 2022	FR (RE2)	59	More details about the team and structure of the startup
	30 May 2022	FR (RE2)	48	Validate and confirm the findings obtained from previous interviews

Table 9 Summary of interview findings and integrative memos

First interview findings	Memo	Second interview findings	Memo	Third interview
Lisy Information began to be obtained related to the management of business processes carried out at Lisy. Some of the information obtained is existing functions and departments, organizational views on processes, process changes that occur, organizational prioritization, process documentation, technology roles, process evaluation, investor relations, to employee <i>reward</i> systems	detailed information has not been obtained regarding the division of teamwork, the development of organizational structure, the development of process documentation, and the pattern of innovation carried out	Lisy does not prioritize levels in organizational structure, but efficiency The driver of Lisy's activities and innovations is customer needs Lisy already has process documentation in the form of SOPs for <i>stable processes</i>, such as employee payroll processes, attendance, and fulfilling certain customer requests	The organizational structure becomes clearer, but more information is needed with respect to the characteristics of organizational structure Further information is needed with respect to process documentation, SOP and how it is improved Finally, we need to explore the culture within the startup	The organizational structure is more fluid than traditional companies in general. This can be seen from the various levels of leaders, some are C-level, head, and even just managers In relation to process documentation, it is known that Lisy already has SOPs that explain activity points and their visualizations in the form of flowcharts. The SOP continues to be improved as they carry out their activities. Another important thing obtained is related to the department responsible for managing the process The organizational culture of <i>the Lisy</i> startup began to be revealed in this interview. Just like <i>startups</i> in general, flexibility is the word that best describes the culture

Table 9 (continued)

First interview findings	Memo	Second interview findings	Memo	Third interview
Rise Overview of the divisions in Rise's organizational structure. In general, the division is almost the same as that of Lisy. Both have operations, <i>IT or tech</i>, and sales <i>or</i> business development <i>divisions</i> information was obtained that Rise is a logistics <i>startup</i> certified by ISO 9001 and 27,001. Therefore, the processes in Rise have been well documented, considering that the process approach is one of the principles of ISO 9001. However, process evaluation is not done routinely, but only when necessary. In addition, the processes in Rise are common processes found in the logistics industry	a lot of information that has not been obtained, such as organizational culture, process management details, process evaluation, process improvement, relationships with stakeholders, and the correlation of organizational strategy with processes	More detailed information was obtained regarding the management of processes in Rise. This information includes team division, conditions that drive the emergence of a process, process documentation, project management, process evaluation, and process improvement. In addition, aspects of organizational culture, human resource conditions, <i>performance appraisal</i>, and <i>stakeholders</i> are also found in this iteration	It is not yet known the people within the startup. How much do the employees understand process management? It is also necessary to explore Rise's organizational culture. More needs to be gathered regarding the process documentation	several new concepts and categories were obtained that complemented the previous findings. The new concepts and categories are related to a <i>startup</i> culture that is open to change, training facilities provided to employees, the background of ISO certification, the absence of special functions that manage processes, and procedures for restructuring. In addition to exploring additional information, this interview validated the findings obtained

Appendix C: Axial coding Lisy and Rise

Lisy concept or category	Main category	Rise concept or category	Main category
Process discovery by understanding the problem and its root cause Brainstorming with users Shadowing method and quantified survey	Process origin	Process determination based on logistic common practice No business process architecture Reactive preparation of business processes	Process origin
Process documentation Business as usual has been documented SOP in the form of bullet points and basic flowcharts Documentation changes are made on the go SOP evolves following the problems being addressed SOP per department SOP changes by supervisor, manager, head, and C-level Process hierarchy in the form of a flowchart	Process management	Innovation based on customer demands	Process management
Multiple participants are required for unique customer requests		Project initiatives New features based on customer needs Business development	
Dividing big problems into small problems (Divide and conquer framework)		Inter-departmental collaboration	
Product management Project (initiative) Worked with a cross-departmental team Improvement by the product team		Project management Application development project led by product manager Product management is more structured Non-application project led by the initiator	
Operational department processes The Ops team division based on order type Dedicated management by field operations On-call managed by operations manager Project managed by planner ops Workflow documentation every time there is a new shipper Preparatory meeting every time there is a new project CEO is still doing technical work RACI matrix for each process		Order management Regular order management (routine) Bidding scheme Project-based order management	

Lisy concept or category	Main category	Rise concept or category	Main category
Process formalization triggers			
Customer demand			
Potential strategic initiatives			
Root cause analysis	Process improve- ment	Not all divisions carry out process improvements	Undisciplined process improvement
Sprint retrospective		Continuous improvement has not been done	
Post mortem analysis		Decentralized process improvement	
Pillar of customer-obsessed to find out the satisfaction of customer problem-solving	Process evalu- ation	No process evaluation Goal-oriented	Improper pro- cess evaluation
Net promoter score (NPS)		No process assessment	
Weekly review (meeting)		Measurement of the results of a process	
		Focus on business KPIs	
Frequency of evaluation as needed		Reactive process evalu- ation	
SOP review every month		Process evaluation fre- quency as needed	
KPI from shipper SLA		Monthly review	
Process evaluation and improvement occur very quickly			
Open communication	Culture	Open communication	Culture
Open mentality to any input			
Open to team ideas		Freedom of change	
Growth mindset			
Learning by doing			
Internal promotion of potential junior employees			
Adaptive			
Agile		Fast-paced	
Three main functions	Organi- zational structure	Divisions	Organizational structure
Commercial (product sales to custom- ers)		Operations	
Sales (search customer requests)		Transport sourcing	
Transporter management (search for suppliers/transporters)		Control tower	
Operational		Field operations	
Control tower and customer service as monitoring team and front liner		Proof of delivery (POD)	
Proof of delivery as an inspector of travel documents		Transport management	
Field operation as the coordinator of the loading process at the shipper's location		Service Excellence	
Allocation as a truck unit finder		Data	
Head-manager-employee		Customer experience	
Export–import department		Operational excellence	
Management of meeting customer needs		Finance	
Product		Account management	
Product		Business development	
		IT	
		Product	
		Engineer	
		HR	

Lisy concept or category	Main category	Rise concept or category	Main category
Search for problems and solutions			
Product manager understands customer problems and needs			
Product managers have their own domain experts			
Engineering as a platform developer			
Back office			
Finance			
People operations			
Incomplete leadership position		Leadership level inconsistency	
Reporting to the position above it		Effect of job scope	
Number of people that can be coordinated		Promotion	
Capability or capacity issue		Initial position offering	
Head-manager-supervisor-employee		Team naming inconsistency	
The structure evolves according to job specialization needs		No special function that manages the process	
Rapid change in organizational structure			
Outside pressure on customer demand			
Inside pressure on high costs and resources			
Chief process officer		Three times organizational restructuring in a year	
Analyst (associate) in product function		HR and finance approval	
Analyst (associate) attached to a certain position		Informal C-level approval	
Project management office (PMO) ensures efficient problem solving			
Operational excellence in commercial function			
Manager in back office function			
External consultant		Team formation follows the leader specialization	
Business advisor		C-level	
Filling the leadership vacancies		CEO	
		CTO	
		CFO	
Basic training	Employee training	Product knowledge training	Training
Skill training		Soft skills training initiated by each division	
Specific training		Employee onboarding	
Based on the competency mapping framework			
Services expansion to the export–import sector based on customer needs	Services	Software as a service (SaaS)	Services
70% of orders via WhatsApp		Trucking marketplace	
Innovations in ordering processes, unit search, order tracking, and assistance supported by technology			

Lisy concept or category	Main category	Rise concept or category	Main category
Objective key result (OKR) Have not implemented OKR in 2019 The organization is still too small OKR each team member OKR is derived from the organization's vision and mission Performance development and discussion every semester based on OKR	Performance measure- ment	Index performance plan (IPP)	Performance appraisal
Reward for achieving milestones (ad hoc)	Rewards	Index performance review (IPR)	Rewards
Reward for performance achievement (routine)		KPI achievement	
The sales department has a reward system from the beginning		Certain project	
Reward in the form of playing together at the beginning of development			
Engineering department has a reward system in the first 6 months			

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